David Zambrano

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EDUCATION

Florida Polytechnic University, Lakeland, FL

Bachelor of Science in Computer Engineering – *GPA: 3.88* May 2026

Master of Science in Electrical Engineering – *GPA: 3.5* Graduating May 2027

PROFESSIONAL EXPERIENCE

Undergraduate & Graduate Research Assistant | *Bin Tarik Nanophotonics Group* May 2025 – Present

- Conducted end-to-end tapeout for several designs including a novel optical ring resonator & all-optical logic gates (NOT, NAND, NOR), operating at the center of the optical C-band range (1550nm).
- Designed & fabricated subwavelength-engineered moiré ring resonator for PUF applications.
- Utilized Lumerical & Flexcompute's Tidy3D for device geometry specification and electromagnetic simulation, and KLayout for GDSII layout for fabrication on 220nm Silicon-on-Insulator (SOI) process by Applied Nanotools.

R&D Intern | *Measutronics Corporation* May 2024 – Aug 2024

- Created an alarm system for dredging vehicles using programmable logic controller (PLC) technology, to warn of hazardous/restricted zones not immediately visible by dredging vehicle operators.
- Utilized microcontroller programming skills to create a robust system, complete with GUI for user configuration.

DESIGN PROJECTS

Subwavelength-engineered Moiré Ring Resonator | *Bin Tarik Nanophotonics Group* Aug 2025 – Present

- Designed a novel ring resonator with width-modulated sidewalls to induce azimuthal perturbation enabling tunable spectra, mode splitting, and localized resonances for filtering, nonlinear optics, and optical computing.
- Electromagnetic simulations demonstrate unique disorder-induced signatures, suitable for hardware security applications in the way of a Physical Unclonable Function (PUF).

Cascadable All-Optical Logic Gates | *Bin Tarik Nanophotonics Group* May 2025 – Aug 2025

- Designed an all-optical inverter from microring resonators, utilizing destructive interference effects for logic inversion for a passive configuration without the use of electro-optic or thermo-optic tuning.
- Electromagnetic simulations confirm transmission inversion, from which 2-input NAND and NOR gates are approximated. SEM imaging confirms CMOS fabrication compatibility thanks to intentional design.

RISC-V 32I Processor | *Florida Polytechnic University* Mar 2025 – May 2025

- Constructed a single-core RV32I processor in the Intel Quartus Prime FPGA SDK, implementing 10 arithmetic and branch logic instructions.
- Validated on DE-10 Lite FPGA board through purpose written assembly program to recursively calculate the Fibonacci sequence for 16 iterations.

LEADERSHIP EXPERIENCE & AWARDS

Leadership: Phoenix Autonomy Electrical Lead (Jan 2026 – Present), Phoenix Autonomy Battery Lead (Aug 2025 – Dec 2026), Poly Concert Band Vice Captain (Aug 2024 – Present), Violet Voices Choir Bass Section Lead (Aug 2024 – May 2025), Presidential Ambassador (Jan 2023 – May 2025).

Awards: 1st Place Research Presentation – Second Annual Poly Research Day, 1st Place Paper Presentation – Second Annual Student Research Symposium IEEE Student Branch, IEEE-HKN Honor Society Member, President's & Provost's List, Saddle Creek Logistics Scholar, Florida Bright Futures Academic Scholar.

SKILLS

Languages: C, C#, C++, Python, Embedded C, MIPS assembly, Verilog & SystemVerilog, Java

Software: Lumerical, Luceda, Mentor Graphics Pyxis, Klayout, Tidy3D, MATLAB, gdsfactory, Electric, KiCad, STM32CubeIDE, Microchip Studio, Intel Quartus Prime, Multisim, LTspice, Visual Studio, SolidWorks (CSWA).

Hardware: GDSII Layout, PCB Layout, FPGA programming, microcontroller embedded programming, real-time operating systems (RTOS), oscilloscope & function generator proficiency, soldering, 3D modeling & printing.

Relevant Coursework: Optoelectronics, VLSI Design, Embedded Operating Systems, Computer Architecture and Organization, Microprocessors, Object Oriented Programming, Data Structures and Algorithms, Digital Logic Design.